

Higher School of Economics

Calculus 2

Author: Andrey Mazhuga

Converted to LaTeX by Sergey Zyablyuk

Calculus 2

1 Lecture 1

Series (Prelim)

Definition 1.1. A function of the form

$$f : \mathbb{N} \rightarrow \mathbb{R}, \quad n \mapsto a_n,$$

where $\mathbb{N}$ is the set of all positive integers and $\mathbb{R}$ is the set of real numbers, is called a *sequence* (of real numbers).

Notations.

$$(a_n)_{n \in \mathbb{N}} = (a_n)_{n=1}^{\infty} = (a_n) = (a_1, a_2, \dots, a_n, \dots).$$

For example,

$$(n^2)_{n \in \mathbb{N}} = (n^2)_{n=1}^{\infty} = (n^2) = (1^2, 2^2, 3^2, \dots, n^2, \dots) = (1, 4, 9, \dots, n^2, \dots)$$

is the sequence of squares.

Definition 1.2. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Then the n -th *partial sum* (of $(a_n)_{n=1}^{\infty}$) is defined to be

$$S_n = \begin{cases} a_1, & n = 1, \\ S_{n-1} + a_n, & n \geq 2. \end{cases}$$

For example, if $\left(\frac{1}{n}\right)_{n=1}^{\infty}$ is a given sequence, then

$$S_1 = \frac{1}{1} = 1,$$

$$S_2 = S_1 + \frac{1}{2} = 1 + \frac{1}{2} = \frac{3}{2},$$

$$S_3 = S_2 + \frac{1}{3} = \frac{3}{2} + \frac{1}{3} = \frac{11}{6},$$

and, in general,

$$S_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}.$$

Definition 1.3. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Then the *series*

$$\sum_{n=1}^{\infty} a_n$$

is defined to be

$$\sum_{n=1}^{\infty} a_n \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} S_n,$$

where S_n is the n -th partial sum of $(a_n)_{n=1}^{\infty}$.

Notation. Instead of $\sum_{n=1}^{\infty} a_n$ one can use

$$a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots,$$

the expanded form.

Examples

1)

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \cdots + \frac{1}{n} + \cdots$$

is the *harmonic series*.

2)

$$\sum_{n=1}^{\infty} \alpha \rho^{n-1} = \alpha + \alpha \rho + \alpha \rho^2 + \cdots + \alpha \rho^{n-1} + \cdots$$

is the *geometric series*.

3) Every real number $x \in [0, 1)$ can be written as

$$x = 0.a_1 a_2 \dots a_n \dots, \quad a_n \in \{0, 1, \dots, 9\}.$$

This notation is a shorthand for the series

$$x = \sum_{n=1}^{\infty} \frac{a_n}{10^n}.$$

For instance,

$$0.2282828 \dots = \frac{2}{10} + \frac{2}{10^2} + \frac{8}{10^3} + \frac{2}{10^4} + \frac{8}{10^5} + \frac{2}{10^6} + \cdots.$$

4)

$$\sum_{n=1}^{\infty} (-1)^{n+1} = 1 - 1 + 1 - 1 + \cdots + (-1)^{n+1} + \cdots.$$

Definition 1.4. Let $(a_n)_{n=1}^{\infty}$ be a sequence. Then the series $\sum_{n=1}^{\infty} a_n$ is called

- *convergent* (to S) if

$$\lim_{n \rightarrow +\infty} S_n = S \in \mathbb{R}, \quad S_n = a_1 + \cdots + a_n$$

is the n -th partial sum;

- *divergent to $\pm\infty$* if

$$\lim_{n \rightarrow +\infty} S_n = \pm\infty;$$

- *divergent* if $\lim_{n \rightarrow +\infty} S_n$ does not exist.

Examples of convergent series

1) If $|\rho| < 1$, then

$$\begin{aligned} \sum_{n=1}^{\infty} \alpha \rho^{n-1} &\stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n \alpha \rho^{k-1} \\ &= \lim_{n \rightarrow +\infty} \alpha \frac{1 - \rho^n}{1 - \rho} = \alpha \frac{1 - 0}{1 - \rho} = \frac{\alpha}{1 - \rho} \in \mathbb{R}. \end{aligned}$$

2)

$$\begin{aligned}\sum_{n=1}^{\infty} \frac{1}{n(n+1)} &\stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{k(k+1)} \\ &= \lim_{n \rightarrow +\infty} \left(1 - \frac{1}{n+1}\right) = 1.\end{aligned}$$

This is a telescoping series.

3)

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{k^2} = \frac{\pi^2}{6}.$$

This is not trivial; we will prove it later in our course.

4)

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{(-1)^{k+1}}{k} = \ln 2.$$

Examples of series divergent to $+\infty$

1)

$$\sum_{n=1}^{\infty} 1 \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \sum_{k=1}^n 1 = \lim_{n \rightarrow +\infty} n = +\infty.$$

2) If $\rho > 1$ and $\alpha > 0$, then

$$\sum_{n=1}^{\infty} \alpha \rho^{n-1} \stackrel{\text{def}}{=} \lim_{n \rightarrow +\infty} \alpha \frac{1 - \rho^n}{1 - \rho} = +\infty.$$

3)

$$\sum_{n=1}^{\infty} \frac{1}{n} = \lim_{n \rightarrow +\infty} \sum_{k=1}^n \frac{1}{k} = +\infty.$$

We will prove this later.

Examples of divergent series

1)

$$\sum_{n=1}^{\infty} (-1)^{n+1} = \lim_{n \rightarrow +\infty} \sum_{k=1}^n (-1)^{k+1},$$

where

$$\sum_{k=1}^n (-1)^{k+1} = \begin{cases} 1, & n \text{ is odd,} \\ 0, & n \text{ is even.} \end{cases}$$

Hence the limit does not exist.

2)

$$\sum_{n=1}^{\infty} \sin(n)$$

is divergent: the limit $\lim_{n \rightarrow +\infty} \sum_{k=1}^n \sin(k)$ does not exist. We will prove this later.

Claim 1.1. Changing a finite number of terms in a series does not affect its convergence (although it may change the sum of the series).

Proof. Let $\sum_{n=1}^{\infty} a_n$ be the initial series and let $\sum_{n=1}^{\infty} b_n$ be obtained from it by changing a finite number of terms. Then it is clear that there exists $N \in \mathbb{N}$ such that

$$\forall n \in \mathbb{N}, \quad n \geq N \implies a_n = b_n.$$

Therefore, for every $n \geq N$,

$$A_n - B_n = A_N - B_N,$$

where

$$A_k = \sum_{i=1}^k a_i, \quad B_k = \sum_{i=1}^k b_i$$

are the k -th partial sums. Thus

$$\lim_{n \rightarrow +\infty} (A_n - B_n) = A_N - B_N \in \mathbb{R},$$

and the right-hand side does not depend on n . By the limit laws for sequences (see Calculus 1), it follows that either both limits $\lim A_n$ and $\lim B_n$ exist, or neither exists. Hence the convergence of the series is unchanged. $\square$

Claim 1.2 (Cauchy's criterion for series). Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ converges} \right) \iff \left(\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m, k \in \mathbb{N}, m \geq N(\varepsilon) \implies \left| \sum_{n=m+1}^{m+k} a_n \right| < \varepsilon \right).$$

Proof. By [Definition 1.4](#),

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \exists \lim_{N \rightarrow +\infty} A_N, \quad A_N = \sum_{n=1}^N a_n.$$

By Cauchy's criterion for sequences (see Calculus 1), this is equivalent to

$$\forall \varepsilon > 0 \exists N(\varepsilon) \in \mathbb{N} : \forall m, k \in \mathbb{N}, m \geq N(\varepsilon) \implies |A_{m+k} - A_m| < \varepsilon.$$

Since

$$\begin{aligned} A_{m+k} - A_m &= (a_1 + \cdots + a_m + a_{m+1} + \cdots + a_{m+k}) - (a_1 + \cdots + a_m) \\ &= a_{m+1} + \cdots + a_{m+k} = \sum_{n=m+1}^{m+k} a_n, \end{aligned}$$

the claim is proved. $\square$

Claim 1.3 (The negation of the Cauchy criterion). Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ diverges} \right) \iff \left(\exists \varepsilon > 0 : \forall N \in \mathbb{N} \exists m \geq N \exists k \in \mathbb{N} : \left| \sum_{n=m+1}^{m+k} a_n \right| \geq \varepsilon \right).$$

Proof. Exercise (or one can see that it follows directly from [Claim 1.2](#)). $\square$

Claim 1.4. The following statements hold:

- 1) $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges;
- 2) $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

Proof. **1)** To prove this fact we use [Claim 1.2](#). Let $\varepsilon > 0$ be arbitrary. Set

$$N(\varepsilon) = \left\lceil \frac{1}{\varepsilon} \right\rceil.$$

For every $m \geq N(\varepsilon)$ and every $k \in \mathbb{N}$,

$$\begin{aligned} \left| \sum_{n=m+1}^{m+k} \frac{1}{n^2} \right| &= \frac{1}{(m+1)^2} + \cdots + \frac{1}{(m+k)^2} \\ &\leq \left(\frac{1}{m} - \frac{1}{m+1} \right) + \cdots + \left(\frac{1}{m+k-1} - \frac{1}{m+k} \right) \\ &= \frac{1}{m} - \frac{1}{m+k} < \frac{1}{m} \leq \frac{1}{N(\varepsilon)} \leq \varepsilon, \end{aligned}$$

where we used

$$\frac{1}{n^2} \leq \frac{1}{n(n-1)} = \frac{1}{n-1} - \frac{1}{n}.$$

Therefore, by [Claim 1.2](#), $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges.

2) We use [Claim 1.3](#). Let $\varepsilon = \frac{1}{2}$. Let $N \in \mathbb{N}$ be arbitrary and set $m = N$, $k = N$. Then

$$\begin{aligned} \left| \sum_{n=m+1}^{m+k} \frac{1}{n} \right| &= \sum_{n=N+1}^{2N} \frac{1}{n} \\ &= \frac{1}{N+1} + \frac{1}{N+2} + \cdots + \frac{1}{N+N} \\ &\geq \underbrace{\frac{1}{2N} + \frac{1}{2N} + \cdots + \frac{1}{2N}}_{N \text{ times}} = N \cdot \frac{1}{2N} = \frac{1}{2} = \varepsilon. \end{aligned}$$

Therefore, by [Claim 1.3](#), $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges. □

Claim 1.5 (Necessary condition for convergence). Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ converges} \right) \implies \left(\lim_{n \rightarrow +\infty} a_n = 0 \right).$$

Note. [Claim 1.4\(2\)](#) shows that the converse does not hold. Indeed,

$$\lim_{n \rightarrow +\infty} \frac{1}{n} = 0, \quad \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges.}$$

Proof. This follows directly from [Claim 1.2](#) if we set $k = 1$. □

Claim 1.6. The series

$$\sum_{n=1}^{\infty} \sin(nx)$$

diverges for all $x \notin \pi\mathbb{Z}$.

Proof. Suppose $\sum_{n=1}^{\infty} \sin(nx)$ converges. Then, due to [Claim 1.5](#),

$$\lim_{n \rightarrow +\infty} \sin(nx) = \lim_{n \rightarrow +\infty} \sin((n+1)x) = 0.$$

Since

$$\sin((n+1)x) = \sin(nx) \cos x + \cos(nx) \sin x$$

and $\sin x \neq 0$ (because $x \notin \pi\mathbb{Z}$), it follows that

$$\lim_{n \rightarrow +\infty} \cos(nx) = 0.$$

But this contradicts the identity

$$\sin^2(nx) + \cos^2(nx) = 1.$$

□

Definition 1.5. Let

$$\sum_{n=1}^{\infty} a_n = S$$

be a convergent series, and let

$$S_n = \sum_{k=1}^n a_k$$

be its n -th partial sum. Then, for every $n \in \mathbb{N}$,

$$S = S_n + r_n,$$

where the sequence $(r_n)_{n=1}^{\infty}$,

$$r_n = \sum_{k=n+1}^{\infty} a_k,$$

is called the *tail* (or the *remainder*) of the series $\sum_{n=1}^{\infty} a_n$.

Claim 1.7. Let $\sum_{n=1}^{\infty} a_n$ be a series. Then

$$\left(\sum_{n=1}^{\infty} a_n \text{ converges} \right) \implies \left(\lim_{n \rightarrow +\infty} r_n = 0 \right).$$

Proof. Let $\sum_{n=1}^{\infty} a_n = S \in \mathbb{R}$. Then

$$r_n = S - S_n \quad \forall n \in \mathbb{N}.$$

Therefore,

$$\lim_{n \rightarrow +\infty} r_n = \lim_{n \rightarrow +\infty} (S - S_n) = S - \lim_{n \rightarrow +\infty} S_n = S - S = 0.$$

□

Claim 1.8 (Linear combination of series). Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be two convergent series. Then, for every $\alpha, \beta \in \mathbb{R}$, the series

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n)$$

is also convergent and

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha \sum_{n=1}^{\infty} a_n + \beta \sum_{n=1}^{\infty} b_n.$$

This series is the linear combination of the series $\sum a_n$ and $\sum b_n$ with coefficients α and β .

Proof. Since $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent, for some $a, b \in \mathbb{R}$ we have

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow +\infty} A_n = a, \quad A_n = \sum_{k=1}^n a_k,$$

and

$$\sum_{n=1}^{\infty} b_n = \lim_{n \rightarrow +\infty} B_n = b, \quad B_n = \sum_{k=1}^n b_k.$$

Therefore, by the limit laws for sequences (see Calculus 1),

$$\begin{aligned} \alpha a + \beta b &= \alpha \lim_{n \rightarrow +\infty} A_n + \beta \lim_{n \rightarrow +\infty} B_n \\ &= \lim_{n \rightarrow +\infty} (\alpha A_n + \beta B_n) \\ &= \lim_{n \rightarrow +\infty} \left(\sum_{k=1}^n (\alpha a_k + \beta b_k) \right). \end{aligned}$$

By [Definition 1.3](#), this is precisely

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha a + \beta b.$$

□

Claim 1.9 (Grouping theorem). Let

- a) $\sum_{n=1}^{\infty} a_n$ be a series;
- b) $(K_n)_{n=1}^{\infty}$ be a sequence of positive integers such that

$$K_1 = 1, \quad K_i < K_{i+1} \quad \text{for every } i \in \{1, 2, \dots\};$$

- c) $\sum_{n=1}^{\infty} b_n$, where

$$b_n = a_{K_n} + \dots + a_{K_{n+1}-1}, \quad n \in \mathbb{N},$$

be a *grouped series*.

Thus b_n is the n -th block and $K_{n+1} - K_n$ is the size of the n -th block. Then:

1)

$$\left(\sum_{n=1}^{\infty} a_n = \alpha \in \mathbb{R} \right) \implies \left(\sum_{n=1}^{\infty} b_n = \alpha \right).$$

2) If

$$\lim_{n \rightarrow +\infty} a_n = 0, \quad \exists m \in \mathbb{N} \forall n \in \mathbb{N} : K_{n+1} - K_n < m, \quad \sum_{n=1}^{\infty} b_n = b \in \mathbb{R},$$

then

$$\sum_{n=1}^{\infty} a_n = b.$$